Objective:

The objective of Module A of this project is to write a single-threaded application that will be then converted to multi-threaded version in Module B.

Project Description:

You are given an input image file with name of "input.pgm". A PGM (Portable GrayMap) image file begins with a simple header that describes the image format, dimensions (i.e., width and height), and pixel value range.

For example:

P5
640 480
255

In this example:
P5 is the image format
640 is width
480 is height
255 is maximum pixel value.

The given image file is a greyscale image. You need to convert the image to black and white form. A greyscale image is converted to black and white form by updating each value of pixel by using following logic.

If (pixel value>=threshold)

{set pixel value to 255}

Else

{set pixel value to 0}.

Given Files:

1. skelecton_project2moduleA.c
2. input.pgm
3. Makefile

Required:

1. Complete the skeleton code by writing all the missing parts (indicated by ToDos) in skelecton_project2moduleA.c
2. After completing the missing parts, run the executable file of skelecton_project2moduleA with following command.

    ./ skelecton_project2moduleA input.pgm output_image.pgm 128

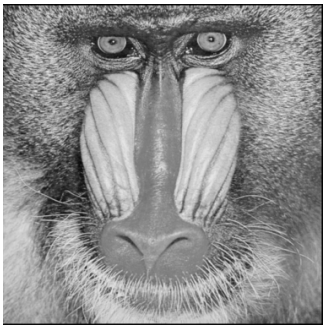
Deliverables.

1. **Complete Source Code:**
    a. **Evaluation:** 70 points for the working code. No points for incomplete code.
2. **output_image.pgm**
    a. **Evaluation:** 30 points for uploading the black and white image file generated by the completed code.

Submission Guidelines:

Place all your files including Makefile in a single folder, name it as first_student_id_first_student_name_second_student_id_second_student_name, zip it, and submit.

Sample Input:



Sample Output:

